

Factors and Multiples

Factor

If two counting (also called natural) numbers are multiplied together, then each one of these numbers is called a factor of the product so obtained. Thus, in the mathematical sentence $7 \times 9 = 63$, 7 and 9 are **factors** and 63 is the **product**. You can also see that $63 \div 7 = 9$ and $63 \div 9 = 7$ and there is no remainder in each case. Therefore, we can also say,

If a counting number divides another number exactly (without any remainder), then the divisor is a **factor** of the dividend.

Example 1: List all the factors of the following numbers.

(a) 16

(b) 17

(a) $16 = 1 \times 16$

$= 2 \times 8$

$= 4 \times 4$

$\therefore$ The factors of 16 are 1, 2, 4, 8 and 16.

(b) $17 = 1 \times 17$

$\therefore$ The factors of 17 are 1 and 17.

From the above example, it is clear that,

1. Every number has at least two factors—1 and the number itself.
2. All the factors of a number are either less than or equal to the number.

Prime Number

A prime number is any natural number **which has exactly two factors**—1 and the number itself.

For example, $13 = 1 \times 13$.

There are no other factors of 13, so 13 is a prime number.

Composite Number

Every natural number except 1 that has **more than two different factors** is called a composite number.

Tips: The number 1 is neither prime nor composite. It is not a prime because it does not have exactly two factors and it is not composite because it does not have more than two factors.

Sieve of Eratosthenes

A Greek mathematician, Eratosthenes, devised a method to find out prime numbers between 1 and 100.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

The steps used in this method are as follows:

Step 1: Write all the numbers from 1 to 100 in a table as shown.

Step 2: Cross out 1 as it is not a prime number.

Step 3: Circle 2 and cross out multiples of 2.

Step 4: Circle 3 and cross out multiples of 3.

Step 5: Circle 5 and cross out multiples of 5.

Step 6: Circle 7 and cross out multiples of 7.

Circle all the numbers that are not crossed out.

The circled numbers are all the prime numbers between 1 and 100.

How many prime numbers have you found between 1 and 100?

Tips: This method is called ‘The Sieve of Eratosthenes’ because factors are used to sift out certain numbers which are multiples just as a sieve sifts out chaff from wheat.

Example 2: Tell whether each of the numbers 48 and 23 is composite or prime.

$$\begin{array}{l} 48 = 1 \times 48 \\ \quad = 2 \times 24 \\ \quad = 3 \times 16 \\ \quad = 4 \times 12 \\ \quad = 6 \times 8 \end{array} \left. \vphantom{\begin{array}{l} 48 = 1 \times 48 \\ \quad = 2 \times 24 \\ \quad = 3 \times 16 \\ \quad = 4 \times 12 \\ \quad = 6 \times 8 \end{array}} \right\} \begin{array}{l} \text{10 factors, therefore,} \\ \textbf{Composite} \end{array}$$
$$23 = 1 \times 23 \} \begin{array}{l} \text{exactly 2 factors,} \\ \text{therefore, } \textbf{Prime} \end{array}$$

Twin Primes

Twin Primes are the prime numbers that differ by 2.

For example, 3 and 5, 5 and 7, 11 and 13, etc. are twin primes.

Co-primes

Two natural numbers are said to be **co-primes**, If they have only 1 as a common factor.

For example, 2 and 3, 5 and 7, 3 and 4, 4 and 9, etc. are co-primes.

Prime Factorisation

Any composite number can be written as a product of prime numbers. An expression written as a product of its prime factors is called the **prime factorization** of that number.

Example 3: Find the prime factorization of 120.

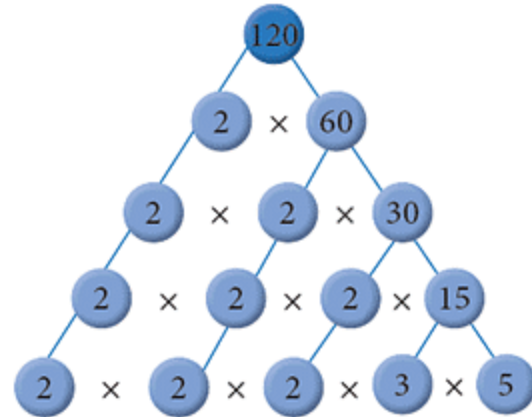
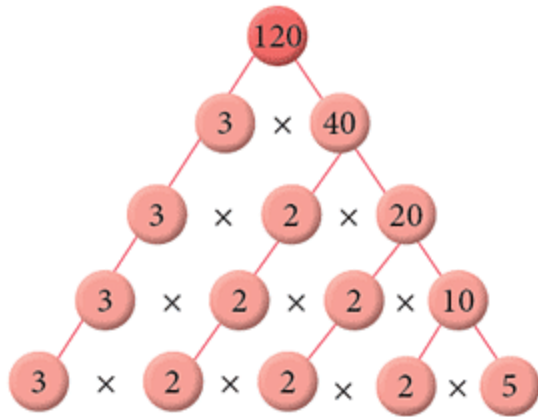
Method 1: Factor Tree

Step 1: Write the number to be factorised at the top.

Step 2: Choose any pair of factors as branches. If either of these factors is composite, factorise again.

Step 3: Choose a pair of factors of each composite number and continue the branches for prime factor(s).

Step 4: Keep factorising till you have a row of prime factors.



Conventionally, the prime factors are written in increasing order.

$\therefore$ The prime factorisation of $120 = 2 \times 2 \times 2 \times 3 \times 5$.

Method 2: Continuous Division

Step 1: Start dividing by the smallest prime factor until you cannot divide any more.

Step 2: Continue dividing by the next higher prime factor.

Step 3: Repeat the process till you get 1.

2	120
2	60
2	30
3	15
5	5
	1

$\therefore$ The prime factorisation of 120 is $2 \times 2 \times 2 \times 3 \times 5$.

Tips: You can have more factor trees for the number 120. Try! But at the end, the prime factors will be the same.

Highest Common Factor (HCF)

If a number is a factor of two or more whole numbers, it is called a common factor of these numbers.

For example, $4 \times 2 = 8$ and $4 \times 5 = 20$. So, 4 is a common factor of 8 and 20.

The **Highest Common Factor (HCF)** of two or more numbers is the greatest of the common factors. It is the greatest number which divides the given numbers exactly.

You can find the HCF of two or more numbers by different methods, as follows:

Method 1: By Listing Factors

Example 4: Find the HCF of 12 and 40.

Step 1: List all the factors of each number.

Factors of 12: 1, 2, 3, 4, 6 and 12

Factors of 40: 1, 2, 4, 5, 8, 10, 20 and 40.

Step 2: Identify common factors.

The common factors are 1, 2 and 4.

Step 3: Identify largest number in the list of common factors.

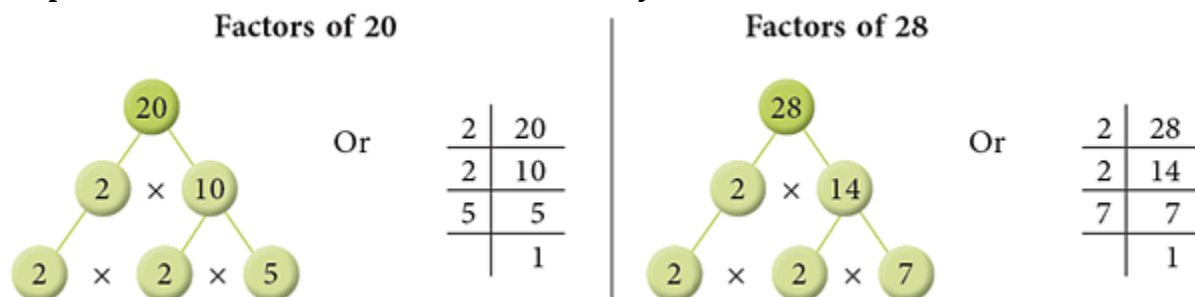
Out of these common factors, 4 is the largest.

∴ HCF of 12 and 40 is 4.

Method 2: By Prime Factorisation

Example 5: Find the HCF of 20 and 28.

Step 1: List all the factors of each number by either method.



Step 2: Identify the common factors.

$$20 = 2 \times 2 \times 5$$

$$28 = 2 \times 2 \times 7$$

Step 3: HCF is the product of common prime factors.

∴ HCF of 20 and 28 = $2 \times 2 = 4$.

Tips: You can get the prime factorisation by making factor trees or by continuous division. However, continuous division is the preferred method.

Method 3: By Division Method

Example 6: Find the HCF of 12, 18 and 42.

Step 1: Divide the numbers by common factors only.

Step 2: Stop dividing when there are no more common factors except 1.

Step 3: Find the product of common factors. 2, 3 and 7 do not have any common

factor except 1, so stop division.

2	12, 18, 42
3	6, 9, 21
	2, 3, 7

$\therefore$ HCF of 12, 18 and 42 = $2 \times 3 = 6$.

Tips: You can use any of the three methods listed above to find the HCF of two or more numbers.

HCF of Large Numbers

It is easier to find the HCF of large numbers by division method.

Example 7: Find the HCF of 217 and 686.

Step 1: Divide the greater number by the smaller number.

Step 2: Make the remainder the new divisor and the original divisor, the new dividend. Repeat the process until you get 0 as the remainder.

Step 3: The last divisor is the required HCF. Here, the last divisor is 7. So, the HCF of 217 and 686 is 7.

$$\begin{array}{r} 217 \overline{) 686} \left(3 \right. \\ \underline{- 651} \\ 35 \overline{) 217} \left(6 \right. \\ \underline{- 210} \\ \text{Last divisor} \rightarrow 7 \overline{) 35} \left(5 \right. \\ \underline{- 35} \\ 0 \end{array}$$

In case of three numbers, first find the HCF of any two numbers.

Then, take the last divisor of the two numbers and the remaining number and find their HCF.

Example 8: Find the HCF of 275, 525 and 830.

First, we take 275 and 525.

Now, find the HCF of 25 and 830.

$$\begin{array}{r}
 275 \overline{) 525} \begin{array}{l} 1 \\ - 275 \\ \hline 250 \end{array} \begin{array}{l} 275 \begin{array}{l} 1 \\ - 250 \\ \hline 25 \end{array} \begin{array}{l} 250 \begin{array}{l} 10 \\ - 250 \\ \hline 0 \end{array} \end{array}
 \end{array}$$

HCF of 275 and 525 is 25.

∴ The HCF of 275, 525 and 830 is 5.

$$\begin{array}{r}
 25 \overline{) 830} \begin{array}{l} 33 \\ - 75 \\ \hline 80 \end{array} \begin{array}{l} 5 \\ - 75 \\ \hline 5 \end{array} \begin{array}{l} 25 \begin{array}{l} 5 \\ - 25 \\ \hline 0 \end{array} \end{array}
 \end{array}$$

HCF of 25 and 830 is 5.

Multiple

A **multiple** of a whole number is the product of the number and any counting number.

We can state some multiples of 2 and 11 as:

$$\begin{array}{l}
 1 \times 2 = 2 \\
 2 \times 2 = 4 \\
 3 \times 2 = 6 \\
 4 \times 2 = 8 \\
 5 \times 2 = 10
 \end{array}
 \left. \vphantom{\begin{array}{l} 1 \times 2 = 2 \\ 2 \times 2 = 4 \\ 3 \times 2 = 6 \\ 4 \times 2 = 8 \\ 5 \times 2 = 10 \end{array}} \right\} \text{Multiples of 2}$$

$$\begin{array}{l}
 1 \times 11 = 11 \\
 2 \times 11 = 22 \\
 3 \times 11 = 33 \\
 4 \times 11 = 44 \\
 5 \times 11 = 55
 \end{array}
 \left. \vphantom{\begin{array}{l} 1 \times 11 = 11 \\ 2 \times 11 = 22 \\ 3 \times 11 = 33 \\ 4 \times 11 = 44 \\ 5 \times 11 = 55 \end{array}} \right\} \text{Multiples of 11}$$

From the above discussion, we can observe these properties of multiples:

1. Every number is a multiple of 1.
2. The smallest multiple of any number (other than zero) is the number itself.
3. Every multiple of a number is greater than or equal to that number.
4. The number of multiples of a given number is infinite.

Example 9: Find the first five multiples of 3.

$$\left. \begin{array}{l} 1 \times 3 = 3 \\ 2 \times 3 = 6 \\ 3 \times 3 = 9 \\ 4 \times 3 = 12 \\ 5 \times 3 = 15 \end{array} \right\} \text{Multiples of 3}$$

$\therefore$ The first five multiples of 3 are **3, 6, 9, 12 and 15**.

Least Common Multiple (LCM)

If a number is a multiple of two or more numbers, it is called a **common multiple** of the numbers.

For example, $2 \times 9 = 18$.

$\therefore$ 18 is a common multiple of 2 and 9.

The smallest number (other than zero) that is a multiple of two or more counting numbers is the **least common multiple (LCM)** of the numbers. It is the least number which is divisible by all the given numbers.

You can find the LCM of two or more numbers by different methods, as follows:

Method 1: By Listing the Multiples

Example 10: Find the LCM of 3, 6 and 12.

Step 1: List some multiples of each number.

The multiples of 3 are 3, 6, 9, 12, 15, 18, 21, **24**, 27, 30, 33, **36**, 39, ...

The multiples of 6 are 6, 12, 18, **24**, 30, **36**, 42, 48, ...

The multiples of 12 are 12, **24**, **36**, 48, ...

Step 2: Identify common multiples.

Some common multiples of 3, 6 and 12 are 24, 36, ...

Step 3: Pick up the least common multiple from the list of common multiples.

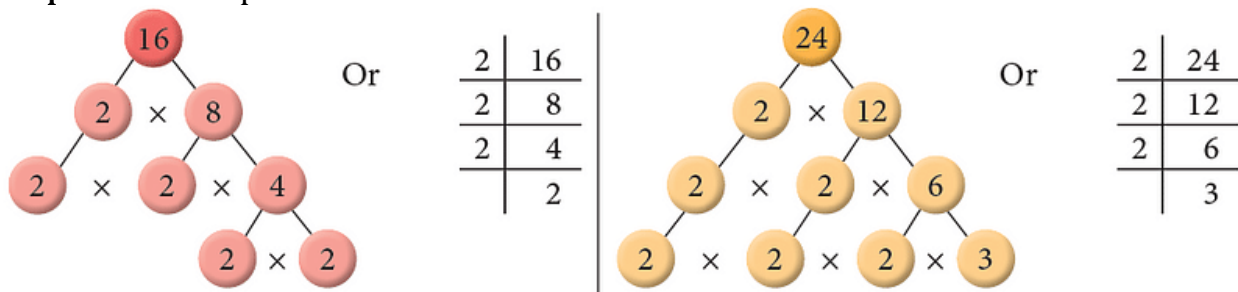
Out of these common multiples, 24 is the least.

$\therefore$ The least common multiple (LCM) of 3, 6 and 12 is **24**.

Method 2: By Prime Factorisation

Example 11: Find the LCM of 16 and 24.

Step 1: Find the prime factorisation of each number.



Step 2: Identify the common factors.

$$16 = 2 \times 2 \times 2 \times 2$$

$$24 = 2 \times 2 \times 2 \times 3$$

Step 3: Multiply the common factors and the extra factors to get the LCM.

$$\therefore \text{LCM} = \underbrace{2 \times 2 \times 2}_{\text{Common factors}} \times \underbrace{3 \times 2}_{\text{Extra factors}} = 48.$$

$\therefore$ LCM of 16 and 24 is **48**.

Method 3: By Division Method

Example 12: Find the LCM of 20, 28 and 40.

Step 1: Divide the numbers by a prime factor common to at least 2 of the given numbers. Bring down the number as such, if it is not completely divisible by the prime factor.

Step 2: Stop dividing when there is no further common factor except 1.

Step 3: Find the product of the numbers in the left column and the last remainders.

2	20,	28,	40
2	10,	14,	20
2	5,	7,	10
5	5,	7,	5
	1,	7,	1

$\therefore$ LCM of 20, 28 and 40 = $2 \times 2 \times 2 \times 5 \times 7 = \mathbf{280}$.

Divisibility Rules

- If a number is a divisor of a second number, then the number is divisible by the first.

For example, since 3 is a divisor of 6, so 6 is divisible by 3.

- If a number is not prime, then it would be divisible by a number smaller than itself. We can find out by actual division whether a number is divisible by another number or not.

For example, if we divide 392 by 7, we find that the remainder is 0, so 392 is divisible by 7. In some cases, the process can be shortened by applying certain tests on the digits of the number.

$$\begin{array}{r} 56 \\ 7 \overline{) 392} \\ \underline{- 35} \\ 42 \\ \underline{- 42} \\ 0 \end{array}$$

Below, we give some such tests:

1. Divisibility by 2

A number is divisible by 2 if its units digit is divisible by 2.

So, a number is divisible by 2 if its units digit is 0, 2, 4, 6 or 8. A number ending in 1, 3, 5, 7 or 9 is not divisible by 2.

For example, the numbers 120, 3172, 234, 81396, 105098 are all divisible by 2 while numbers like 13, 287, 335, 7091, 28469 are not divisible by 2.

2. Divisibility by 3

A number is divisible by 3 if the sum of its digits is divisible by 3.

For example, the number 384 is divisible by 3 and the sum of its digits ($3 + 8 + 4$) is 15, which is divisible by 3.

Similarly, the number 217095 is also divisible by 3 as the sum of its digits ($2 + 1 + 7 + 0 + 9 + 5$) is 24, which is divisible by 3.

The number 839 is not divisible by 3 as the sum of its digits ($8 + 3 + 9$) is 20, which is not divisible by 3.

3. Divisibility by 5

A number is divisible by 5 if its units digit is 0 or 5.

For example, the numbers like 205, 3075, 2370, 15000 are all divisible by 5. Numbers like 27, 3189, 200053 are not divisible by 5 as they do not end in 0 or 5.

4. Divisibility by 9

A number is divisible by 9 if the sum of its digits is divisible by 9.

For example, the number 7326 is divisible by 9, because $7 + 3 + 2 + 6 = 18$ and 18 is divisible by 9. The number 27041 is not divisible by 9, because the sum of its digits $2 + 7 + 0 + 4 + 1 = 14$, which is not divisible by 9.

5. Divisibility by 10

A number is divisible by 10 if its units digit is 0.

For example, the numbers 100, 8390, 9500, 5401,000 are all divisible by 10. Numbers like 209, 703001, 28597, 4385 are not divisible by 10 as their units digit is not 0.

6. Divisibility by 11

A number is divisible by 11 if the difference of the sum of its digits in odd places and the sum of its digits in even places (starting from the units place) is either 0 or divisible by 11.

Study the following table.

Numbers	Sum of the Digits in Odd Places	Sum of the Digits in Even Places	Difference
3465	$5 + 4 = 9$	$6 + 3 = 9$	$9 - 9 = 0$
6457	$7 + 4 = 11$	$5 + 6 = 11$	$11 - 11 = 0$
95986	$6 + 9 + 9 = 24$	$8 + 5 = 13$	$24 - 13 = 11$
280929	$9 + 9 + 8 = 26$	$2 + 0 + 2 = 4$	$26 - 4 = 22$

So, the numbers 3465, 6457, 95986 and 280929 are all divisible by 11.

Below, we give rules to test the divisibility of a number by composite numbers like 4, 6, 8, 12 and 25:

1. Divisibility by 4

A number is divisible by 4 if the number formed by its digits in tens and units places is divisible by 4.

For example, in the number 80372, the number 72 formed by the tens and the units digit 2 is divisible by 4. As you can check this number is divisible by 4.

2. Divisibility by 6

A number is divisible by 6 if it is divisible by both 2 and 3, i.e., it should be an even number and the sum of its digits should be divisible by 3.

For example, the number 68370 is divisible by 6. It is an even number and also sum of its digits ($6 + 8 + 3 + 7 + 0$) is 24, which is divisible by 3.

3. Divisibility by 8

A number is divisible by 8 if the number formed by the digits at the

hundreds, tens and units place is divisible by 8.

Consider the number 207608.

The number formed by the digits at the hundreds, tens and units place, i.e., 608 is divisible by 8.

$$\begin{array}{r} 8 \overline{) 207608} \\ \text{Q } 25951 \end{array}$$

$\therefore$ This number is divisible by 8.

Since 705 is not divisible by 8, so the number 8705 is not divisible by 8.

$$\begin{array}{r} 8 \overline{) 8705} \\ \text{Q } 1088 \text{ R}1 \end{array}$$

4. Divisibility by 25

A number is divisible by 25 if the number formed by the digits at the tens and units places is divisible by 25.

For example, in the numbers 8750, 23275, 8926825, we observe that 50, 75 and 25, i.e., the numbers formed by the digits at the tens and units places, are divisible by 25. So, all these numbers are divisible by 25.

$$\begin{array}{r} 25 \overline{) 8750} \\ \text{Q } 350 \end{array}$$

From the above rules, we observe that:

A number is divisible by another number if it is divisible by its co-prime factors. Thus, number divisible by 2 and 5 will also be divisible by 10.

Example 13: Check the divisibility of 19440 by 18.

The co-prime factors of 18 are 2 and 9.

Therefore, to check the divisibility of 19440 by 18, we check whether the given number is divisible by 2 and 9 both.

19440: It is an even number and so is divisible by 2.

Sum of the digits = $1 + 9 + 4 + 4 + 0$

$= 18$, which is divisible by 9, so

19440 is also divisible by 9.

Thus, 19440 is divisible by 2×9 , i.e., 18.